Health & MedTech Hackathon 2025

Software Track — Problem Statements

Department of Computer Science & Engineering

All problem statements require purely software-based solutions — no hardware or medical devices required.

The following seven problem statements challenge participants to build impactful digital health solutions addressing critical gaps in India's healthcare ecosystem — from AI-assisted diagnosis and telemedicine to medicine safety, mental health, and frontline worker support. Teams must select one problem statement and deliver a working software prototype by the hackathon deadline.

Problem Statement 1 of 7

HC-01 · Software for Speech & Language Therapy Clinical Services

Problem Description

- Helps speech and language therapists manage therapy services digitally.
- Enables patient assessment and diagnosis recording.
- Allows creation of personalized therapy plans.
- Tracks patient progress over time.
- Stores session records securely.
- Generates digital reports automatically.
- Includes interactive therapy exercises.
- Supports AI-based speech analysis for improvement tracking.
- Makes therapy more organized, accessible, and effective.

Problem Statement 2 of 7

HC-02 · Emergency Care System – Real – Time Ambulance Connectivity

Problem Description

- Connects ambulances, doctors, and hospitals in real time.
- Shares live patient data (vital signs, condition, medical history).
- Tracks ambulance location using GPS.
- Allows doctors to monitor patients remotely.
- Enables doctors to guide paramedics before hospital arrival.
- Helps hospitals prepare equipment and treatment in advance.
- Reduces emergency response time.
- Improves survival rates and patient outcomes.

Problem Statement 3 of 7

HC-03 · Telemedicine & Remote Patient Monitoring Platform

Problem Description

- Provides video and audio consultation between patients and doctors.
- Supports text-based asynchronous consultation for low-bandwidth areas.
- Allows doctors to issue digital prescriptions.
- Enables patients to log vital data (BP, glucose, SpO2, weight).
- Includes remote patient monitoring module.
- Sends alerts to doctors when vitals cross safe thresholds.
- Designed for low connectivity and affordability.
- Improves healthcare accessibility in underserved areas.

Problem Statement 4 of 7

HC-04 · Decentralized Organ Donation & Transplant Matching System

Problem Description

- Creates an immutable donor-recipient registry using blockchain.
- Ensures transparency and prevents corruption.
- Stores verified donor and recipient records securely.
- Uses smart-contract-based automated matching.
- Matches based on medical compatibility (blood type, HLA markers).
- Prioritizes urgency automatically.
- Eliminates human bias and administrative delays.
- Improves fairness and efficiency in organ allocation.

Problem Statement 5 of 7

HC-05 · Medicine Authenticity Verification & Drug Interaction Checker

Problem Description

- Allows users to scan medicine barcode/QR code for authenticity verification.
- Connects to blockchain or government drug database.
- Detects counterfeit medicines.
- Provides drug interaction checker.
- Generates AI-based interaction warnings.
- Shows severity ratings and alternative suggestions.
- Stores user medication history.
- Automatically flags harmful drug combinations.

Problem Statement 6 of 7

HC-06 · Community Health Worker (ASHA/ANM) Digital Support Tool

Problem Description

- Digitizes beneficiary registration.
- Tracks pregnancy and child immunization schedules.
- Records anemia and malnutrition screening data.
- Automates follow-up scheduling.
- Works offline and syncs when internet is available.
- Provides central dashboard for monitoring.
- Supports voice-based data entry in regional languages.
- Reduces data loss and reporting delays.
- Improves efficiency of frontline health workers.

Problem Statement 7 of 7

HC-07 · Remote Surgical Assistance using AR & Real-time Low-Latency Streaming

Problem Description

- Addresses the shortage of specialist surgeons in rural and underserved areas.
- Enables remote specialists to guide local surgeons during complex procedures.
- Uses Augmented Reality (AR) for real-time visual guidance.
- Allows specialists to draw annotations on the surgeon's live field of view.
- Supports placement of 3D markers on specific anatomical structures.
- Requires ultra-low latency video streaming for real-time interaction.
- Ensures precise spatial mapping so markers stay fixed on body parts.
- Improves surgical accuracy and reduces procedural risks.
- Expands access to expert surgical care in remote locations.

Guidelines: Teams of 2–4 members. Open-source tools, public datasets, and free-tier APIs are permitted. Final submission must include a working demo, GitHub repository, and a 5-minute presentation. Judging criteria: Innovation (25%), Technical Complexity (25%), Healthcare Impact (25%), and Demo Quality (25%).